

DIPOLE TYPE BY LENGTH

STEP 0 — always start here λ = wavelength (m), l = antenna rod length (m), $c=3\times 10^8$ m/s

$\lambda = \frac{c}{f}$ (m)	ratio $\frac{l}{\lambda} \rightarrow$	pick row below
l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R, \theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\ \Omega$
$= 1$	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\ \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
$\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$, $\vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta$ (W/m ²)
l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k=2\pi/\lambda$ (rad/m); $\eta_0=120\pi \approx 377\ \Omega$.
Max @ broadside $\theta_{\text{max}} = \pi/2$; D exact $D=1.5$ (1.76 dB)
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S=0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.
Normalized pattern <i>dimensionless</i> $F(\theta) = S(\theta)/S_{\text{max}}$
Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$, $\cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$, $S_0 = \frac{15 I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$
$D=1.64$, $R_{\text{rad}}=36.5\ \Omega$, $\theta_{\text{max}} = \pi/2$.
Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$, <i>shortcut form</i>
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$
$D=1.64$, $R_{\text{rad}}=73\ \Omega$, $\theta_{\text{max}} = \pi/2$.

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion <i>multiply by $\pi/180$ or $180/\pi$</i>
$\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$
° rad sin θ cos θ sin ² θ cos ² θ
0 0 0 1 0 1
30 $\pi/6$ 1/2 $\sqrt{3}/2$ 1/4 3/4
45 $\pi/4$ $\sqrt{2}/2$ $\sqrt{2}/2$ 1/2 1/2
60 $\pi/3$ $\sqrt{3}/2$ 1/2 3/4 1/4
90 $\pi/2$ 1 0 1 0
180 π 0 -1 0 1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

Setup N identical elements along axis, spacing d
elem pos $z_i = id$, $i = 0, \dots, N-1$; $A_i = a_i e^{j\psi_i}$
Equal phase ($\psi_i=0$) $\Rightarrow$ broadside beam ($\theta_{\text{max}} = \pi/2$).
Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$.
Pattern multiplication S_e = single element pattern
$S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$
Array factor (general) $k=2\pi/\lambda$
$F_a(\theta) = \left \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$
Uniform $A_i=1$, equal phase <i>closed form</i>
$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$, $\gamma = kd \cos \theta$
Peak: $F_a^{\text{max}} = N^2$ at $\gamma=0$ (broadside, $\theta = \pi/2$). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$.
HPBW (broadside, uniform) <i>use Nd, not $(N-1)d$</i>
$\beta \approx 0.88 \frac{1}{\sqrt{Nd}}$ (rad) = $50.8^\circ \cdot \frac{1}{\sqrt{Nd}}$
Grating lobes appear if $d > \lambda$ (broadside). Avoid.

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c=3\times 10^8$ m/s
I_0 — peak current (A)
$k=2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377\ \Omega$
$S(R, \theta)$ — pwr density (W/m ²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
a — coef. on θ^2 in Gaussian
I_0 — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
ξ — rad. efficiency; $G = \xi D$
$D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIS / dB

P_t — power transmitted (W)
P_r — power received (W)
G_t — transmit gain (linear)
G_r — receive gain (linear)
$S_r = G_t P_t / (4\pi R^2)$ (W/m ²)
$P_r = G_t G_r (\lambda / 4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB = $\times 2$, 10 dB = $\times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m ²)
$A_e = \lambda^2 D / (4\pi)$ (m ²)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p / \lambda^2$ (unitless)
$\beta \approx 0.88 \lambda / \ell$ (rect/sq.)
$\beta \approx 1.02 \lambda / d$ (circular)
$2A_p \Rightarrow 2D$, $\beta / \sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd \cos \theta$
$F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
$F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a / N^2$
$\beta \approx 0.88 \lambda / (Nd)$ (broadside)

BEAM PARAMETERS — β , Ω_p , D

Normalize <i>always start here</i>
$F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)
Beamwidth β at threshold X
Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)
Threshold $X=10^{\text{dB}/10}$ θ_X (Gauss. short-cut)
HPBW (−3 dB) 0.5 $\sqrt{\frac{\ln 2}{a}}$
−6 dB 0.25 $\sqrt{\frac{\ln 4}{a}}$
−10 dB 0.1 $\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$ any X $F(\theta_X) = X$
Pattern solid angle Ω_p <i>recognize pattern, look up</i>
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$ (sr)
No ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$.
θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.
Pattern $F(\theta)$ Ω_p (sr)
Isotropic ($F=1$) $4\pi \approx 12.57$
Hertzian sin ² θ $8\pi/3 \approx 8.38$
Half-wave dipole ≈ 7.66
Narrow Gauss $e^{-a\theta^2}$ $\pi/a = \pi\theta_{1/2}^2 / \ln 2$
2-axis HPBW (aperture) $\beta_{xz} \cdot \beta_{yz}$
Other (use integral) numerical

TI-36X Pro: [2nd] [d/dx] for $\int_0^\pi$, RAD mode, multiply by 2π .

Directivity D <i>always $D=4\pi/\Omega_p$; common shortcuts below</i>
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
WORD TRAP: “direction of max radiation” = angle θ_{max} (rad/deg). “directivity” = unitless number D . Different things!
Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p / \lambda^2$. 2-axis HPBW: $D \approx 4\pi / (\beta_{xz} \beta_{yz})$. Circular: $D \approx 4\pi / \beta^2$.
Gain (if asked) $\xi = \text{rad. eff.}$, $0 < \xi \leq 1$
$G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.

COMMON DIPOLE PARAMETERS

Type	l	$D = D_{\text{dB}}$	R_{rad} (Ω)
		G (lossless)	
Isotropic	N/A	1	0
Hertzian	$< \lambda/50$	1.5	1.76
Half-wave	$\lambda/2$	1.64	2.15
Monopole	$\lambda/4$ (gnd)	1.64	2.15
Full-wave	λ	2.41	3.82

Half-wave: $P_{\text{rad}} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.
Lossless antenna ($\xi=1$) $\Rightarrow G=D$. So in Friis, use G_t, G_r = these D values (e.g. half-wave $G=1.64$).

ANTENNA AREAS (A_e, A_p)

Effective area <i>antenna's RX cross section; D from BEAM PARAMS or table above</i>
$A_e = \frac{\lambda^2 D}{4\pi}$ (m ²)
Physical cross section <i>wire dipole, diameter d; l see table above</i>
$A_p = l d$ (m ² , any dipole)
Inverse: D from A_e
$D = \frac{4\pi A_e}{\lambda^2}$ (unitless)
Hertzian special case $D=1.5$
$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi^2} \approx 0.119 \lambda^2$ (m ²)
Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim \lambda$.

FRIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F=1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G = 10^{G_{\text{dB}}/10}$.

3. Get G_t, G_r by antenna type:

- named dipole $\rightarrow$ COMMON DIPOLE PARAMS ($G=D$ lossless)
- aperture / dish $\rightarrow$ APERTURE ANTENNAS ($G=D \approx 4\pi A_p / \lambda^2$)
- arbitrary pattern $F(\theta) \rightarrow$ BEAM PARAMS ($G = \xi \cdot 4\pi / \Omega_p$)
- given in dB $\rightarrow$ dB $\leftrightarrow$ LINEAR table

4. Plug into Friis.

Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity ($G, D, P_r/P_t, \text{SNR}, \dots$):

	$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$		$X_{\text{lin}} = 10^{X_{\text{dB}}/10}$
dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

NOISE & SNR

Noise power <i>thermal noise into matched receiver</i>
$P_N = k_B T_{\text{sys}} B$ (W)
$k_B = 1.38 \times 10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).
Signal-to-noise ratio
$\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs SNR > 10 dB for usable signal.

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m ²); A_e effective area (m ² , $= \lambda^2 D / (4\pi)$); β_{xz}, β_{yz} HPBW in two planes (rad); ℓ, d aperture side / diameter (m).
Directivity (any shape, uniform illum.)
$D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$
Directivity from beamwidths <i>any aperture</i>
$D \approx \frac{4\pi}{\beta_{xz} \beta_{yz}}$ (use rad); circular: $D \approx 4\pi / \beta^2$

HPBW and D by aperture shape *uniform illum.; β in rad*

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x \ell_y$	$\beta_x \approx 0.88 \lambda / \ell_x$ $\beta_y \approx 0.88 \lambda / \ell_y$	$D \approx 4\pi / (\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88 \lambda / \ell$	$D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$
Circular (diam. d)	$\pi d^2 / 4$	$\beta \approx 1.02 \lambda / d$	$D \approx \frac{4\pi}{\beta^2} = \frac{4\pi d^2}{\lambda^2}$

Scaling rules $D \propto A_p / \lambda^2$; $\beta \propto \lambda / \sqrt{A_p}$		
Change	New D	New β
Double area ($A_p \rightarrow 2A_p$)	$D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}} / \sqrt{2}$
Double freq ($\lambda \rightarrow D_{\text{new}} = 4D_{\text{old}}$)	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}} / 2$